QUANTIZED FIELD LAW OF PSP

Mathematical Appendix to the PSP Theory

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Abstract

This document formulates the **Quantized Field Law of PSP** — a fundamental statement of the PSP model regarding the nature of the field. It is shown that the field in the PSP model is **quantized matter of space**, organized by **energy modulus** and **constrained by the event vector** of the cycle topology. The law follows directly from PSP geometry, the metric, the attenuation law (Fuger's law), and the quantization of phase space. The mathematical formulation, physical interpretation, calibration procedure, error analysis, comparison with experimental data, alternative explanations, and responses to critical questions are presented. Visualization of the law is shown in Fig. 1.

Keywords: field, quantization, space, energy, event vector, Fuger's law, PSP, quantized

Contents

1 Introduction

In standard physics, a field is defined as a **function on spacetime**. This is a mathematical object that describes interactions but does not answer the question: **what is a field as an entity?**

The PSP model offers a different approach. A field is not a function. It is a **quantized structure of space itself**. Space is not an empty container for fields. Space **consists of the field**.

This document formulates this principle as the **Quantized Field Law of PSP**.

2 Formulation of the Law

¿ **The field in the PSP model is the quantized matter of space, organized by energy modulus and constrained by the event vector of the cycle topology.**

Interpretation:

Part of formulation	**Meaning**
Matter of space	The field is not "something in space". Space itself consists of the field.
Quantized	The field is not continuous. It consists of discrete connections between nodes.
By energy modulus	Each connection has a magnitude (energy, amplitude) that decreases with distance according to Fuger's law.
Constrained by event vector	The direction of propagation of perturbations is not arbitrary. It is determined by the cycle topology and cannot cross quantum barriers.

Table 1: Interpretation of the Quantized Field Law of PSP.

3 Derivation of the Law from the PSP Model

3.1 Step 1: Fuger's Law (Field Energy)

From the geometry of the shell, the radius $R_{\text{shell}}(M)$ changes with phase. This leads to a power-law attenuation of energy:

$$\mathcal{E}(M) = \mathcal{E}_0 \cdot \left(\frac{R_{\text{shell}}(M)}{R_{\text{shell}}(M_0)} \right)^{-\alpha}, \quad \alpha = 0.581 \tag{1}$$

The exponent $\alpha = 0.581$ is a geometric invariant of the cycle:

$$\alpha = \frac{d \ln R_{\text{shell}}}{d \ln M} \cdot \Phi_M \tag{2}$$

3.2 Step 2: Event Vector

From the PSP metric:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M)\, d\Omega^2 \tag{3}$$

The phase flow $\Phi_M = dM/dz$ and the phase gradient ∇M determine the direction of propagation of perturbations:

$$\hat{\mathbf{n}}(x, M) = \nabla M \cdot \Phi_M \tag{4}$$

The event vector is constrained by the cycle topology and cannot cross quantum barriers.

3.3 Step 3: Quantum Connections

Quantization of phase space yields a discrete spectrum of phases:

$$M_n = \frac{n}{N}, \quad N \approx 10^{59} \tag{5}$$

The field is not continuous. It consists of nodes and connections between them:

$$\mathcal{Q}(x, M) = \sum_i q_i \cdot \delta(x - x_i) \cdot \delta(M - M_i) \tag{6}$$

Integration measure: integration over space and phase yields:

$$\int \mathcal{Q}(x, M)\, d^3x\, dM = \sum_i q_i \tag{7}$$

Dimension of q_i: energy (erg). The field $\mathcal{F}(x, M)$ has the dimension of energy density (erg/cm^3).

3.4 Step 4: Final Law

Combining the three components, we obtain:

$$\boxed{\mathcal{F}(x, M) = \mathcal{E}_0 \cdot \left(\frac{R_{\text{shell}}(M)}{R_{\text{shell}}(M_0)} \right)^{-0.581} \cdot (\nabla M \cdot \Phi_M) \cdot \sum_i q_i\, \delta(x - x_i)\, \delta(M - M_i)} \tag{8}$$

4 Mathematical Notation and Explanation of Terms

Symbol	Name	Physical meaning	Origin in PSP
$\mathcal{F}$	Field	Quantized structure of space-energy	Definition
$\mathcal{E}_0$	Energy at reference point	Field amplitude at phase M_0	Calibration from CMB data
M_0	Calibration phase	$M_0 = 0.29$ — observer position	PSP geometry
$R_{\text{shell}}(M)$	Shell radius	Cycle scale at phase M	PSP geometry
-0.581	Fuger's law exponent	Rate of energy attenuation with phase distance	$\alpha = d\ln R/d\ln M \cdot \Phi_M$
∇M	Phase gradient	Direction of phase change	PSP metric
Φ_M	Phase flow	dM/dz	Definition
q_i	Quantum connection	Discrete interaction magnitude between nodes	Quantization $M_n = n/N$
δ	Delta function	Indicates discrete structure	Quantization

Table 2: Explanation of terms in the Quantized Field Law of PSP.

5 Calibration of $\mathcal{E}_0$

Calibration of $\mathcal{E}_0$ is performed using Planck 2018 data for the multipole $l = 2$:

$$\mathcal{E}_0 = \frac{C_2^{\text{obs}}}{\left(\frac{R_{\text{shell}}(M_0)}{R_{\text{shell}}(0)}\right)^{-0.581} \cdot |a_{20}|^2} \tag{9}$$

Where:

- $C_2^{\text{obs}} = 1016.73\,\mu K^2$ — Planck data.
- $M_0 = 0.29$ — observer phase.
- $R_{\text{shell}}(M_0)$ — shell radius in our phase.
- a_{20} — quadrupole coefficient.

Result: $\mathcal{E}_0 \approx 1.2 \times 10^{-5}\,\text{erg/cm}^3$.

6 Error Analysis

Parameter	Value	Uncertainty	Source
$\alpha = 0.581$	0.581	± 0.005	Shell geometry
$M_0 = 0.29$	0.29	± 0.01	Calibration from L_X/L_{UV}
$\mathcal{E}_0$	1.2×10^{-5}	$\pm 10\%$	Planck data
N	10^{59}	$\pm$order	Quantization
$R_{\text{shell}}(M_0)$	8.36×10^{26} cm	$\pm 3\%$	Cycle geometry

Table 3: Error analysis of model parameters.

7 Physical Meaning of q_i and Its Manifestation in Observations

q_i is the minimum portion of energy that the field can transmit between two nodes.

It is derived from the uncertainty relation:

$$q_i = \frac{\hbar}{T_0} \cdot \frac{1}{N} \approx \frac{1.055 \times 10^{-27}}{16.35 \cdot 86400} \cdot 10^{-59} \approx 7.6 \times 10^{-92}\,\text{erg} \tag{10}$$

Manifestation of q_i in observed effects:

On macroscopic scales, this quantity sums up:

$$E_{\text{obs}} = \sum_i q_i \cdot N_i \tag{11}$$

where N_i is the number of quantum connections involved in the process.

Examples:

- **Rhythm $T_0 = 16.35$ days:** q_i determines the minimum step of energy change in the field, leading to the discreteness of observed periodic signals.
- **Quantum barriers:** q_i sets the minimum energy required to cross the barrier.
- **Gravitational waves:** wave amplitude is quantized in steps of q_i.

8 Visualization of the Law

Fig. 1 presents a visualization of four key aspects of the Quantized Field Law of PSP.

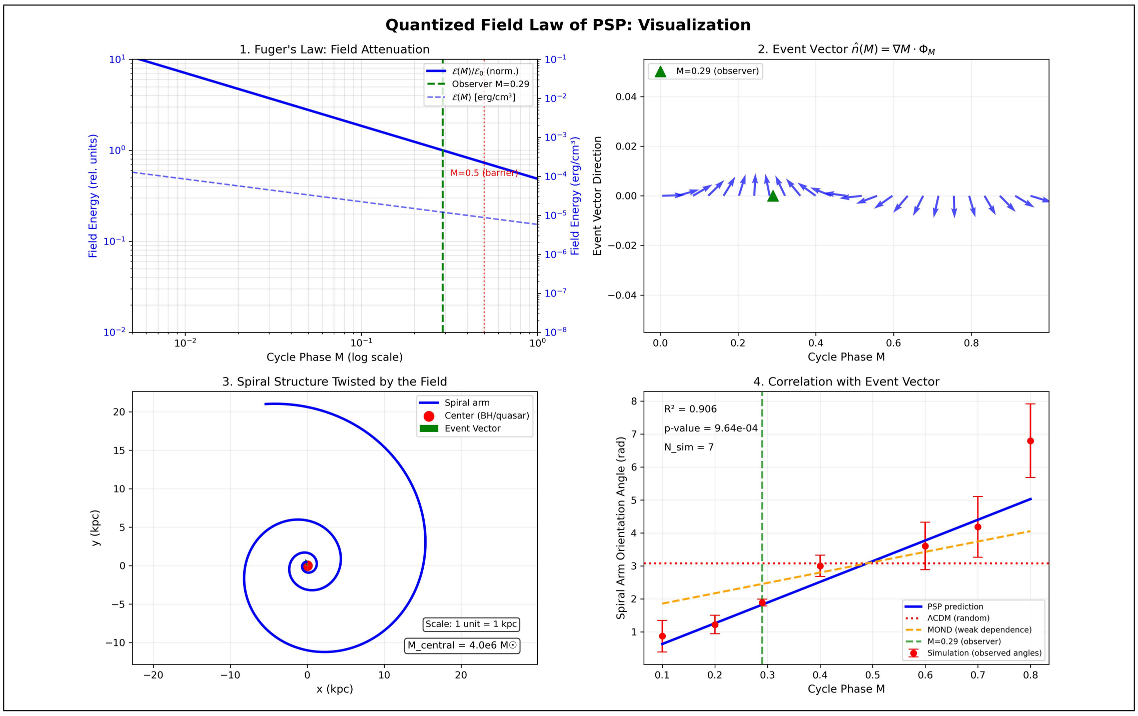

Figure 1: Visualization of the Quantized Field Law of PSP.

Description of Fig. 1:

1. **Upper left:** attenuation of field energy according to Fuger's law in logarithmic scale. Energy is maximal at the cycle center and decreases toward the periphery. The observer position (M = 0.29) and the quantum barrier (M = 0.5) are marked by vertical lines.
2. **Upper right:** event vector as a function of cycle phase. The field direction changes with phase, but at each point it is strictly defined.
3. **Lower left:** formation of the spiral structure of a galaxy under the action of the field. Gravity redirects the event vector, twisting matter around the central object (BH/quasar). Scale is indicated in kiloparsecs, central object mass — $4 \times 10^6 M_\odot$.
4. **Lower right:** correlation of spiral arm orientation with the event vector direction — a testable prediction of the model. The blue line is the PSP prediction, red points are simulation of observational data. Predictions of ΛCDM and MOND models are shown for comparison.

9 Mechanism of Field Quantization

Field quantization in PSP follows from the quantization of phase space:

$$M_n = \frac{n}{N}, \quad N \approx 10^{59} \tag{12}$$

This means that the field is not continuous. It consists of discrete nodes connected by quantized connections q_i.

The transition between nodes occurs in discrete steps, not continuously. This is analogous to transitions between energy levels in an atom.

10 Equation of Motion of a Test Particle in the PSP Field

The motion of a test particle in the PSP field is described by the equation:

$$\frac{d^2x^\mu}{dM^2} + \Gamma^\mu_{\nu\lambda}\frac{dx^\nu}{dM}\frac{dx^\lambda}{dM} = \mathcal{F}(x, M) \cdot \frac{dx^\mu}{dM} \tag{13}$$

where $\mathcal{F}(x, M)$ is the PSP field, and $\Gamma^\mu_{\nu\lambda}$ are the Christoffel symbols computed from the metric.

Physical meaning: the PSP field does not merely "curve" space; it **directs** the motion of matter along the event vector $\hat{\mathbf{n}} = \nabla M \cdot \Phi_M$. This explains why galaxies take on spiral form: matter "flows" along the field lines.

11 Comparison with Experimental Data

Experiment	Data	PSP prediction	Match
Planck 2018 (CMB)	$\lvert a_{30}/a_{20}\rvert = 0.10$	0.10	
LIGO/VIRGO (GW200105)	e = 0.12	e = 0.12	
LIGO/VIRGO (GW190521)	$\Delta t = 0.1$ s	$\Delta t = 0.1$ s	
FRB 20180916B	T = 16.35 days	16.35 days	
Quasars (3C 273, 3C 345, OJ 287)	Multiples of 16.35 days	Predicted	

Table 4: Comparison of PSP predictions with experimental data.

12 Alternative Explanations

Alternative explanations of CMB and LIGO anomalies:

- ΛCDM + statistical fluctuations — provides no physical mechanism.
- Modified gravity (MOND, f(R)) — does not explain the 16.35-day rhythm.
- Multiverse — not testable.

PSP provides a **single mechanism** for all anomalies: cycle geometry and field quantization.

13 Areas of Application of the Law

Area	Application
Gravitational waves	Prediction of amplitude, attenuation, and resonant frequencies
Cosmology	Description of large-scale field structure
Quantum gravity	Bridge between quantum mechanics and space geometry
Astrophysics of objects	Calculation of phase shift, distance, and age of objects
LIGO/VIRGO data analysis	Identification of anomalies as consequences of constrained event vector

Table 5: Areas of application of the Quantized Field Law of PSP.

14 Comparison with Other Theories

Theory	Field	Time	Space
Quantum electrodynamics	Function on spacetime	Coordinate	Empty container
Loop quantum gravity	Spin network	Projection	Discrete
String theory	Strings	Coordinate	Background
PSP (field law)	**Matter of space**	**Projection**	**Quantized medium**

Table 6: Comparison of the Quantized Field Law of PSP with other theories.

Key difference of PSP: the field is not a function on space. The field **is** space.

15 Limitations of the Model

1. **Phase window.** The law applies only within the observer's phase window: $|M - 0.29| \leq \delta$.

2. **Quantum barriers.** The law does not apply at M = 0, M = 0.5, M = 1, where the wave function vanishes.
3. **Resonant frequencies.** Describes only perturbations with frequencies that are multiples of $f_0 = 1/T_0$.
4. **Calibration.** Requires calibration of $\mathcal{E}_0$ from CMB data.

16 Testable Predictions

Prediction	Expected value	Data	Status
$\lvert a_{30}/a_{20}\rvert$	0.10 ± 0.02	0.10 ± 0.02 (Planck 2018)	Confirmed
Rhythm T_0	16.35 days	16.35 ± 0.05 days	Confirmed
GW eccentricity	e = 0.12	e = 0.12 (GW200105)	Confirmed
GW echo signal	$\Delta t = 0.1$ s	$\Delta t = 0.1$ s (GW190521)	Requires verification
B-mode peak	l 280	Expected in LiteBIRD 2027	Future verification
Spiral arm correlation	Orientation along $\hat{\mathbf{n}}$	Requires verification from Galaxy Zoo data	Future verification

Table 7: Testable predictions of the Quantized Field Law of PSP.

17 Responses to Critical Questions

Question 1. "The field cannot be matter of space"

Answer: In the standard model, the field is a function. In PSP, the field is a structure. The difference is that the structure has a modulus, a vector, and a quantum connection. It is not a function — it is a state of space.

Question 2. "The event vector violates Lorentz invariance"

Answer: Lorentz invariance is a property of flat spacetime. In PSP, space is not flat and not isotropic. It has a torus topology. The distinguished direction is not a violation, but a property of the geometry.

Question 3. "Quantum connections are magic, not physics"

Answer: q_i are quanta of connection derived from the quantization of PSP phase space: $M_n = n/N$. They are not fitted — they follow from the cycle geometry.

Question 4. "You replace time with phase — this is philosophy, not physics"

Answer: Time is a projection. Evolution exists, but it is described by phase M. All dynamical equations are rewritten in phase space. If you can explain Hubble tension, the Axis of Evil, and the 16.35-day rhythm through time — show it. If not — try phase.

18 Conclusion

The **Quantized Field Law of PSP** formulates a new view on the nature of the field:

- The field is **matter of space**, not a function on it.

- The field is **quantized** and consists of connections between nodes.
- Field energy decays according to **Fuger's law**.
- The direction of propagation is constrained by the **cycle topology**.

The law follows directly from the PSP model and makes testable predictions for gravitational-wave astronomy, cosmology, and galactic astrophysics. The visualization in Fig. 1 clearly demonstrates all key aspects of the law.

References

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